MFU-LCMP: Land Cover Mapping Using Deep Learning Area for Green Environment

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Abstract / We introduce MFU LCMP, a research focused on evaluating the capability of Deep Learning models for land cover mapping in Northern Thailand, primarily using open source satellite datasets. Unlike previous work that often relied on high resolution imagery from commercial sources, this study explores whether models, if trained exclusively on open datasets on the Hugging Face platform, can generalize to real world imagery of Chiang Rai province without requiring additional new labeling. This research evaluates popular image segmentation architectures, namely U Net, U Net++, and DeepLabv3, with EfficientNet as the backbone, and tests them against images from Google Earth Pro manually labeled by the researchers into 8 classes covering the area in a manner similar to standard remote sensing and computer vision test sets. The MFU LCMP work provides class definitions, evaluation methods, and baseline metrics to support future extension research. We found that although open satellite data is becoming more accessible, the problem of domain shift between global level datasets and the landscape of the Southeast Asian region is understudied and represents a significant performance limitation. The analysis of class specific IoU values, mean mIoU, and confusion matrix helps to illustrate both the potential and the limitations of an open data approach for real world environmental mapping. This work aims to bridge modern Deep Learning techniques with geospatial applications, serving as a reference point for future research in mapping, green space assessment, and sustainable land use planning.

Keywords / remote sensing — semantic segmentation — land cover — U-Net — DeepLabv3 — open data — Chiang Rai

1. Introduction

Satellite and aerial images are like windows that let us witness how our world is changing. We can see cities expanding, farmlands shifting, forests disappearing, and the effects of climate change. This kind of information helps researchers understand what is happening on Earth and also supports real world decisions, from city planning [1] and natural resource management [2], to creating environmental policies [3].

This type of information is called Remote Sensing. Unlike ordinary photos, remote sensing images are captured from satellites or aircraft and can detect light that our eyes cannot see, such as infrared. Because of that, they can reveal details that are invisible to us, such as telling vegetation apart from bare soil or distinguishing water from buildings [4].

However, analyzing this type of data is not simple. Images taken from the same location can look completely different depending on factors such as sunlight, seasons, or the camera's angle. These factors make it difficult to identify and classify features such as roads, trees, or buildings [5]. In the past, traditional methods such as Markov Random Fields (MRF) or Conditional Random Fields (CRF) were used in an attempt to improve segmentation. However, these techniques struggled with the high complexity found in real-world imagery. Today, Deep Learning has demonstrated far superior performance, since it can automatically learn deep and complex features from data. Architectures such as U-Net [6] and DeepLabv3 [7] have become standards for semantic segmentation tasks. Numerous studies have been built upon these architectures, including neural architecture search approaches [8] and hybrid network designs [9, 10]. There are also specialized tools designed to enhance building boundary extraction, such as GeoSeg [11]. However, Deep learning techniques rely on large, high-quality training datasets. High-resolution imagery often comes with a very high cost. For this reason, researchers are turning to use open datasets

For this reason, researchers are turning to use open datasets available on platforms such as Hugging Face. These datasets provide satellite or aerial images along with pixel level labels. Open datasets allow researchers to train Deep Learning models without paying the high cost of collecting and labeling data themselves [12, 13, 14].

Several studies also make their datasets publicly available as open-source resources, focusing on specific geographic areas and land-cover classes. For example, the study by Saito et al. [13] shared a specific dataset in Japan. The dataset utilized remote sensing datasets,

primarily derived from Sentinel-2 and Landsat-8 satellite imagery, to perform land cover mapping and analysis in forest areas [13]. As well as in another study, they also shared a dataset in order to classify forests that are in Hindu Kush area [15].

However, publicly available datasets from previous studies exhibit limitations. The datasets only cover the sepcific areas. They contain only small subsets of high-quality, pixel-level annotations suitable for training deep learning–based segmentation models. Most open datasets still rely on coarse polygons, image-level labels, or incomplete class definitions, which makes them unsuitable for developing detailed land cover maps. Moreover, even with the open data, researchers continue to face a shortage of training samples with labels that are accurate, consistent, and reliable enough for effective model development. [16, 15]

There are also open datasets available on open platforms, such as OpenEarthMap and Google Earth Pro. OpenEarthMap offers global-scale benchmarking satellite imagery. However, OpenEarthMap does not cover the entire planet, particularly in Thailand [12]. In contrast, Google Earth Pro provides satellite imagery that covers almost all areas of the world, including Thailand. Owing to its high spatial resolution and positional accuracy, Google Earth Pro has been widely employed for land-use analysis, mapping, and change detection [17, 18, 19].

When using Google Earth Pro for land cover mapping analysis, the labeling is the costly process. Image labeling is time-consuming and typically requires expert knowledge. Consequently, many studies rely on pre-labeled datasets to reduce both time and expense. However, this practice can introduce a challenge known as domain shift. In remote sensing, this type of shift is often caused by differences in illumination, sensor properties, seasonal effects, or spatial resolution. Deep learning models trained on datasets from one geographic region often exhibit poor performance when applied to another region with substantially different landscape characteristics, vegetation types, and illumination conditions. [20, 21].

Therefore, this study focuses on land cover mapping analysis in the specific area, Chiang Rai, Thailand. A

Europe or North America perform poorly when applied to environments in Thailand [20, 21]. As the landscape , vegetation, and lighting conditions differ significantly between regions.

This situation lead to a new challenge known as Domain Shift, which means differences in data distributions. Most open datasets originate from Europe or North America, where landscapes, vegetation, and lighting conditions differ significantly from those in Northern Thailand. As a result, models trained on data from foreign regions

In some area such as Chiang Rai, Thailand, there is no available open satellite image datasets. The avaliable datasets in the area have to purchase from commercial satellite images from providers such as Maxar or Planet is usually beyond the budget of most research projects [22, 23]. Therefore, this study uses Google Earth Pro imagery as the evaluation dataset because it provides high resolution imagery that covers most areas on Earth and has been proven to have acceptable positional accuracy for research purposes [17, 18, 19].

This research (MFU-LCMP) focuses on addressing this challenge. The goal is to find a cost-effective approach for land-use mapping without the need to create a new labeled dataset.

The objectives of the study are:

(1) To reduce the cost and complexity of creating high-resolution land use maps by using open data from Hugging Face instead of purchasing commercial imagery. This shows that accurate models can be trained without creating new annotations, saving both time and resources.

(2) To identify the most effective Deep Learning architecture. The study compares U-Net, U-Net++, and DeepLabv3 trained with open datasets and evaluates their performance using imagery from Chiang Rai. The assessment focuses on both classification accuracy and efficiency.

Lastly, this research wants to provide evidence that combining open datasets with Deep Learning offers an effective and cost-effective solution for local environmental mapping. The results are shown to support both academic research and sustainable land-use planning in Northern Thailand.

2. Related Work

2.1. From Traditional Methods to Deep Learning Models

In the early days, land cover mapping mostly relied on traditional machine learning methods such as Support Vector Machines (SVM), Decision Trees, and Markov Random Fields (MRF) [7, 24]. These methods worked by designing handcrafted features, for example using specific color or texture patterns to distinguish vegetation from buildings or roads. While effective in simple cases, these features often failed in real-world situations. A model that recognized concrete roads could easily misclassify dirt or shadowed roads, and gray rooftops often looked too similar to asphalt. This issue, known as spectral confusion, was one of the biggest challenges for traditional classifiers [4].

Another difficulty was that these methods did not understand spatial context. They made decisions based mainly on pixel values, without really "seeing" how pixels relate to their surroundings. For instance, MRFs tried to smooth predictions by looking at neighboring pixels, but they were difficult to tune and required heavy computation. SVMs, though powerful in theory, often struggled with the large size and high complexity of satellite imagery, especially when data from different regions looked very different from each other [20, 21].

The arrival of deep learning brought a major change. Instead of relying on handcrafted features, Convolutional Neural Networks (CNNs) learn features directly from data by recognizing patterns, edges, and textures automatically [5]. U-Net [6] introduced an encoder–decoder structure that helped preserve fine details through skip connections. Later, U-Net++ [8] im-

proved this design with more refined pathways, producing smoother boundaries. DeepLabv3 [25] went even further, using Atrous Spatial Pyramid Pooling (ASPP) to capture information at multiple scales, helping the model better understand both small and large objects within one image.

As research continued, hybrid models and transfer learning techniques [9, 10] were developed to make segmentation models more adaptive across different regions and sensors. Yet, deep learning also has its own challenges: it demands a large amount of high-quality labeled data, careful fine-tuning, and often struggles when applied to unfamiliar environments. These ongoing challenges remind researchers that data quality and domain adaptation remain just as important as the models themselves [20, 21].

2.2. Evolution of Datasets: From Natural Images to Satellite Imagery

Before open satellite datasets became widely available, most deep learning models were trained on natural image datasets such as *ImageNet* or *PASCAL VOC*. These datasets focus on common objects like people, vehicles, and animals. Models trained in this context can recognize shapes and textures but lack knowledge about geographic features or land use patterns [5].

In contrast, satellite images represent real geographic surfaces where every pixel corresponds to physical land. They include multiple spectral bands (e.g., visible and infrared) that allow separation of vegetation, soil, water, and built-up areas [4]. However, this data also introduces additional challenges such as cloud obstructions, shadows, seasonal variations, and atmospheric effects. Commercial datasets such as *WorldView* or *SPOT* remain costly and are not openly accessible [23, 22].

In recent years, newly released open datasets have improved research by eliminating the expense of data acquisition. For example, *OpenEarthMap* [12] is a large benchmark dataset containing high-resolution imagery from 97 regions worldwide, with manually created pixel level labels to support model comparison. Other studies focus on region-specific datasets, including work conducted in Japan [13], which utilized Sentinel-2 and Landsat-8 imagery for land cover mapping. Another approach is *SyntheWorld* [14], which uses synthetic images generated from 3D environments (such as GTA-style simulation) to produce large volumes of automatically labeled data. Such datasets are valuable for model development and benchmarking across regions.

However, these datasets are still insufficient for direct application to our target region. Most imagery is collected from Europe or North America, where landscapes, vegetation, and building styles differ significantly from Northern Thailand. This creates a domain shift problem, as models trained on such datasets perform poorly when deployed in unfamiliar environments [20, 21].

Constructing a new dataset specific to Chiang Rai would solve this issue but requires time, expert labor, and budget for image acquisition and manual labeling. Therefore, this research adopts a strategy that integrates two complementary data sources:

Training Data, open datasets hosted on Hugging Face [2], providing a large and diverse global repository of satellite imagery at no cost.

Evaluation Data, real high-resolution imagery from Google Earth Pro, representing our target study area in Chiang Rai Province, which currently lacks publicly labeled datasets [17, 18, 19].

This approach enables direct investigation of the main research question: How well can models trained on global open data generalize and adapt to real-world local imagery in Thailand?

3. Methodology

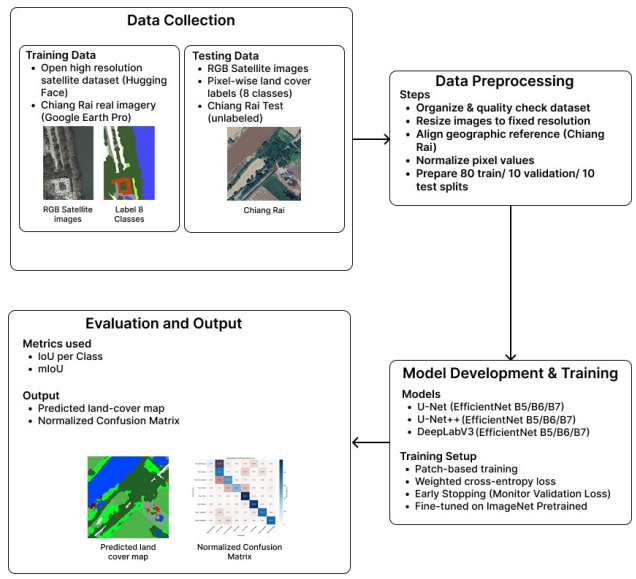


Fig. 1. Workflow of the proposed training process for land cover classification.

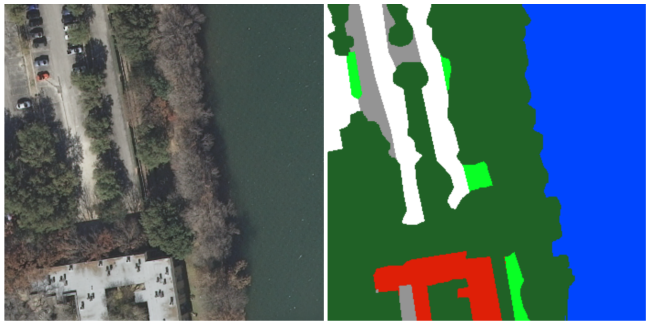
The pipeline begins with high-resolution satellite imagery and corresponding pixel level labels with 8 land-cover classes. The images are used for model training with different backbone architectures, followed by testing and evaluation on unseen data. The final outputs include predicted land-cover maps and confusion matrices that illustrate the model's classification performance.

To conduct land cover mapping in the northern region of Thailand, we utilized an open dataset to train the model. The methodological framework consists of three main stages: data preprocessing, model training, and performance evaluation, as illustrated in the corresponding workflow diagram (Figure 1).

3.1. Data Preprocessing

This study employs opendata sources for training a model:

This study using high resolution satellite imagery from open access datasets published on the Hugging



(A) (B)

Fig. 2. Example (A) land cover class label and (B) corresponding original image.

Face platform to train Deep Learning models for land cover classification [5, 4, 26].

The dataset consists of RGB images with a spatial resolution ranging from 0.25 to 0.5 meters per pixel, with each image sized at either 512×512 or 1024×1024 pixels. Each image is come with pixel level labels that have been labeled by experts, allowing the model to learn detailed spatial characteristics such as texture patterns, object boundaries, and differences among land cover types including water, cropland, forest, roads, and buildings [20, 24].

The data preprocessing stage involves quality inspection, resizing to ensure consistent dimensions, and formatting the data for model input. The dataset is then divided into two subsets: a training set used to train the model, and a validation set used for parameter tuning and to prevent overfitting [27, 28].

Table 1. Class proportions

Class	Pixels	Percent (%)
Bareland	27,755,867	1.673
Grass	346,680,989	20.902
Pavement	307,360,009	18.531
Road	109,205,529	6.584
Tree	314,675,756	18.973
Water	53,158,933	3.205
Cropland	208,876,073	12.594
Building	278,335,850	16.782

The label distribution shows a clear imbalance across the eight classes in the open high resolution satellite dataset Table 1. Grass has the highest proportion of pixels 20.902%, followed by Tree 18.973%, Pavement 18.531%, and Building 16.782%. These dominant classes strongly influence the learning process, allowing the model to more easily capture frequent spatial patterns, such as those found in urban or agricultural areas.

However, Bareland 1.673% and Water 3.205% contain far fewer pixels, meaning the model receives very limited examples of their spatial or spectral characteristics. As a result, these classes are harder to learn and are more prone to misclassification.

This imbalance directly affects semantic segmentation performance. Classes with more samples tend to bias the model toward common patterns, while classes with very few samples often suffer from underfitting and poor generalization. Understanding this distribution is essential for choosing strategies such as data augmentation, class weight adjustments, and interpreting metrics like mIoU and per class accuracy within the context of land cover mapping.

The dataset contains 8 land cover classes [12], defined as follows:

(1) Bareland refers to land without permanent vegetation. It includes soil, sand, gravel, bare farmland, and construction areas.
(2) Grass areas are primarily covered by grass and may contain a few shrubs. These areas are not managed for crop production.
(3) Pavement refers to human altered open surfaces that are not classified as buildings or roads, such as parking lots, sidewalks, and paved courtyards.
(4) Road describes transport surfaces with linear connectivity, including asphalt, concrete, or clearly used dirt roads.
(5) Tree areas are covered with tree canopy, including natural forests, fruit orchards, and roadside trees.
(6) Water represents open surface water bodies such as rivers, lakes, and reservoirs, which may appear dark or reflective.
(7) Cropland refers to land systematically used for crop cultivation, with clear field boundaries that distinguish it from grassland.
(8) Building describes man made permanent structures, including residential, commercial, and industrial buildings, often with geometric shapes and shadows.

The corresponding color scheme used for pixel level annotation is detailed in Table 2.

Table 2. Class Definitions with Colors used in the Training Dataset.

Color (HEX)	Color (RGB)	Class
#800000	[128, 0, 0]	Bareland
#00FF24	[0, 255, 36]	Grass
#949494	[148, 148, 148]	Pavement
#FFFFFF	[255, 255, 255]	Road
#226126	[34, 97, 38]	Tree
#0045FF	[0, 69, 255]	Water
#4BB549	[75, 181, 73]	Cropland
#DE1F07	[222, 31, 7]	Building

The testing dataset is used to evaluate model performance after training is completed. None of the images in this dataset are used for training or validation. which represents typical landscapes in Northern Thailand. Figure 3 shows an example satellite image from the chiang rai area.we label a image by using a CVAT. we determine with the previous mentiomed 8 classes definition. The data are tiled into 512×512 pixel patches covering a variety of environments, including: Urban areas,Agricultural zones ,Forested landscapes.

(A) (B)

Fig. 3. Example (A) land cover class label and (B) corresponding original image.

3.2. Environment Setup

Hardware Specifications

The core hardware setup includes an NVIDIA RTX 4000 SFF Ada Generation GPU (20 GB VRAM), an Intel Core i7-14700 CPU, 32 GB RAM, and a 512 GB NVMe Micron SSD running on Windows 11.

Software and Libraries

The main libraries used are Python 3.13.3, PyTorch 2.6.0 + cu118 (CUDA 11.8), TorchVision 0.21.0, NumPy 2.2.3, OpenCV 4.11.0, Pillow 11.1.0, Rasterio 1.4.3, and Albumentations 2.0.5. Key deep learning dependencies include Segmentation Models PyTorch 0.2.0, EfficientNet PyTorch 0.7.1, Timm 1.0.15, and Pretrainedmodels 0.7.4.

3.3. Modelling (Model Design and Training)

This research focuses on evaluating the performance of three popular Deep Learning architectures for the task of Semantic Segmentation. The models we are assessing are U-Net, U-Net++, and DeepLabv3. We chose these specific architectures because they are widely used and highly effective in analyzing high resolution satellite imagery. Their design is particularly robust, allowing them to handle the complex spatial characteristics often found in these images. This means they are excellent at accurately identifying and segmenting both large objects (like wide farmlands or major roads) and small, intricate objects (such as building outlines or narrow geographical features).

Model Architectures

U-Net is an encoder-decoder architecture designed for precise pixel-level image classification [6]. The encoder section reduces the spatial size of the image to extract deep, essential features, while the decoder section expands the image back to its original resolution to create a full-pixel classification map.

U-Net's key advantage is the use of skip connections to link data from the encoder to the corresponding position in the decoder [6]. This helps the model efficiently merge deep semantic information and spatial information, resulting in increased accuracy for classifying small or thin objects, such as roads or rivers.

U-Net++ is an architecture developed as an extension of U-Net, aimed at resolving the inconsistency of features between the encoder and decoder [8]. It achieves this by adding a structure featuring dense skip connections and intermediate convolution layers within the connection pathways [8].

This structure helps reduce the semantic gap between the features in the two sections, resulting in more consistent and higher quality features being transferred to the decoder. Consequently, the model is able to accurately identify boundaries of objects with complex or discontinuous shapes, such as building edges, farmlands, or other high resolution spatial elements [8].

DeepLabv3 is an architecture designed to handle the diverse range of spatial scales present in satellite imagery [25]. It utilizes the atrous (dilated) convolution technique to expand the receptive field without reducing the spatial resolution of the features [7]. This allows the model to analyze image context simultaneously.

The model also integrates Atrous Spatial Pyramid Pooling (ASPP) to aggregate information from multiple scales [25]. This enables the model to capture characteristics of large areas, such as forests or open spaces, alongside the detection of small objects like waterways and roads. The resulting features are then passed through the decoder to generate a high-resolution, full-resolution segmentation map with high accuracy [25].

Backbone

All three model architectures use EfficientNet as the backbone for feature extraction from satellite imagery [29]. EfficientNet employs the principle of compound scaling, which balances the increase in the network's depth, width, and resolution. This results in high performance efficiency in terms of accuracy relative to the number of parameters used [29]. This research selects three versions of EfficientNet:

- EfficientNet B5: Offers good performance with moderate resource usage [29].
- EfficientNet B6: Has increased depth, making it suitable for data requiring greater contextual interpretation [29].
- EfficientNet B7: Possesses the highest capability, ideal for very high-resolution satellite imagery that demands deep spatial detail capture [29].

The EfficientNet structure consists of an initial convolution layer for extracting basic features, followed by a sequence of MBConv blocks, which include:

(1) depthwise convolution for processing data channel by channel, which reduces computation [29].
(2) pointwise convolution (1x1) for merging data from multiple channels.
(3) Squeeze and Excitation (SE) module for adaptively recalibrating the importance weights of each feature channel [29].

Furthermore, EfficientNet is pretrained on ImageNet [30]. This provides the model with a foundational abil-

ity to recognize basic patterns, such as shapes, textures, surfaces, and object boundaries, which significantly boosts its performance when applied to satellite image classification tasks and substantially reduces the training time [26].

Training Strategy

The training process follows a patch-based approach, where small image patches with pixel level labels from the Hugging Face dataset are used to teach the models how to recognize different land cover types [5, 9]. All models are trained under the same experimental settings to ensure a fair comparison. Each one starts with ImageNet pretrained weights and is fine-tuned on the new dataset. This fine-tuning step allows the models to shift from general visual understanding to recognizing specific satellite features such as textures, edges, and spatial layouts [29, 10, 20].

The dataset is divided into training (80%), validation (10%), and testing (10%) sets to maintain balanced evaluation and ensure reliable performance assessment across all models [4, 21].

Loss Function Design

To improve learning quality, a hybrid loss function is used. It combines:

(1) Weighted Cross Entropy Loss: Helps the model focus more on under represented classes (like Bareland or Water), reducing class imbalance issues.

By combining these three losses, the model learns more effectively. Weighted Cross Entropy ensures each class is fairly represented, Dice Loss sharpens boundaries, and Jaccard Loss improves overall regional accuracy. Working together, they produce clearer and more reliable land cover maps across all eight classes.

3.4. Evaluation Method

To understand how well each model performs the land cover segmentation task, several evaluation tools are used. The main metrics are Intersection over Union (IoU) and Mean Intersection over Union (mIoU), which are commonly relied upon in image segmentation research. A confusion matrix is also included to explore the types of mistakes each model makes. All evaluations are performed on the same test dataset to ensure fairness when comparing the models.

1. Intersection over Union (IoU)

IoU is calculated for each land cover class. It measures how much the predicted region overlaps with the ground truth. The formula is:

$$IoU_i = \frac{TP_i}{TP_i + FP_i + FN_i} \tag{1}$$

where

- TP_i: correctly predicted pixels for class i
- FP_i: pixels incorrectly predicted as class i
- FN_i: pixels belonging to class i that were missed

A higher IoU means the model is better at identifying that class.

2. Mean IoU (mIoU)

mIoU is the average IoU across all eight classes:

$$mIoU = \frac{1}{k}\sum_{i=1}^{k} IoU_i \tag{2}$$

with $k = 8$ in this study. This score gives an overall picture of model performance while treating each class equally, regardless of how large or small the class is in the image.

3. Normalized Confusion Matrix

To gain deeper insight into how each segmentation model performs, a Normalized Confusion Matrix is used. This matrix visualizes how well the model distinguishes between different land cover types, showing both correct predictions and the types of misclassifications that occur. Each row of the matrix represents the actual class, while each column represents the predicted class. The values are normalized so that every row sums to one (or 100%), allowing fair comparison across classes of different sizes scikit-learn. The normalization is calculated as:

$$\hat{CM}_{ij} = \frac{CM_{ij}}{\sum_{j=1}^{k} CM_{ij}} \tag{3}$$

where CM_{ij} is the number of pixels belonging to the i-th true class that were predicted as class j, and $\hat{CM}_{ij}$ is the normalized value for that entry. By normalizing, each row effectively shows the proportion of predictions for a given true class. This makes the results easier to interpret and highlights which categories the model confuses most often. For example, Bareland is sometimes mistaken for Cropland because both share similar soil textures, while Road and Pavement can appear alike due to their comparable colors and linear shapes in high-resolution satellite images. Overall, the normalized confusion matrix offers a clearer, more intuitive view of the model's behavior, helping reveal its strengths and weaknesses in distinguishing between visually similar land cover types

4. Experiments and Results

The optimal number of training epochs was determined using a validation-driven convergence analysis of the mean Intersection over Union (mIoU) curve. Across all architectures, the mean Intersection over Union (mIoU) followed a similar trend, performance increased steadily during the early stages of training especially between epochs 1 and 20 before gradually stabilizing around epochs 25 to 30.

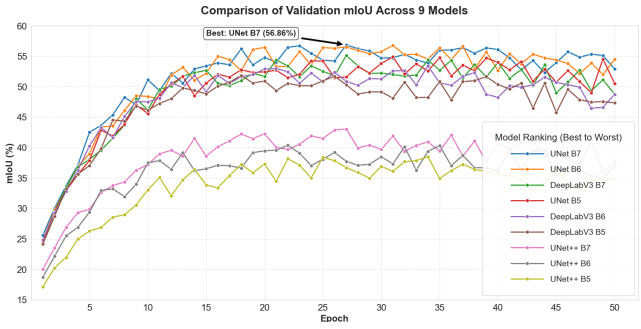


Fig. 4. Comparison of Validation mIoU Across Models

In contrast, U-Net++ (EfficientNet B6) achieved slightly lower overall accuracy and showed more fluctuations at higher epoch counts, even though it still demonstrated early improvement similar to the other models.

Overall, the results indicate that training beyond 30 epochs does not provide meaningful improvements. In some models, signs of overfitting begin to appear once the training extends too long. Considering the stability of validation mIoU, the overall performance across architectures, and the computational cost, this study identifies 30 epochs as the most suitable training duration.

This balance ensures reliable model performance while avoiding unnecessary computation time, an important consideration when developing land cover classification models from high-resolution satellite imagery.

4.1. Quantitative results

Table 3 Quantitative results of land cover segmentation across eight classes using U-Net, U-Net++ and DeepLabv3 with different EfficientNet backbones.

After finalizing the training settings and evaluating all models, the quantitative results are summarized in Table 3. These results clearly show how differently each architecture U-Net, U-Net++, and DeepLabv3 performs when paired with various EfficientNet backbones.

The strongest overall performer is U-Net with EfficientNet B7, achieving an mIoU of 56.53%. This model stands out in classes that have clear and consistent spatial patterns, such as Tree 63.66%, Cropland (65.65%, and Building 78.14%. These classes tend to have stable textures and structure, which makes them easier for the model to learn. However, Bareland 37.45% and Water 36.15% remain difficult, mainly because their appearance often overlaps with other surrounding classes in satellite imagery.

Following closely behind is U-Net with EfficientNet B6 with an mIoU of 55.74%. This model handles spectrally similar classes surprisingly well such as Grass 47.53%, Pavement 47.24%, and Road 57.96% even though these classes often blend together in real world imagery. Its performance on Bareland 39.31% and Water 38.82% is also comparable to B7, though still not as strong as in the more visually distinct categories.

The next model, U-Net with EfficientNet B5, reaches an mIoU of 53.90%. Its performance can be considered moderate, it still does well on Tree, Cropland, and Building, but it struggles more in areas where classes look very similar particularly Grass vs. Cropland and Pavement vs. Roads, which are commonly confused even by human labelings.

DeepLabv3 with EfficientNet B6 follows with an mIoU of 52.26%. Although it performs slightly below the U-Net-based models, it remains effective for clearer classes such as Building and Cropland. Its limitations become more noticeable in classes with irregular patterns , especially Bareland and Water, where textures vary widely across scenes.

The weakest performing model in this study is U-Net++ with EfficientNet B6, which achieved an mIoU of 39.30%. Its accuracy drops significantly in classes like Bareland 17.19%, Water 15.49%, and Cropland 38.97%, likely because the model overfits easily when the dataset contains few and highly diverse samples. Still, it performs reasonably well in the Tree class 53.98%, which provides more consistent visual cues for the model to learn from.

To better understand how each model performs in real world data, beyond the results presented earlier, this study also provides qualitative comparisons. These visual examples include the original satellite images, the ground truth labels, and the predictions generated by each model. As shown in the figure, these comparisons help reveal the segmentation patterns, the continuity of different land cover regions, and the specific areas where each model performs well or struggles.

This is especially noticeable in more complex classes such as Tree, Cropland, and river edge areas, where land cover types often appear very similar.

When comparing the outputs of U-Net with EfficientNet B7 with other strong performers like U-Net B6 and DeepLabv3, clear differences emerge in boundary precision and in how effectively the models separate classes that share similar visual characteristics in real world landscapes. These differences help explain why some architectures consistently achieve higher accuracy than others.

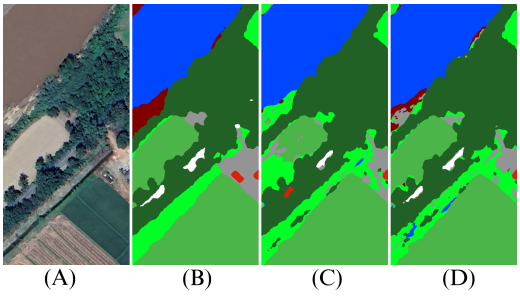


Fig. 5. Example of land cover classification results.(A) Original image,(B) label of this image .(C) U-Net using EfficientNet B7 , and (D) U-Net using EfficientNet B6.

The image sample is located in Chiang Rai Province at latitude 19.9677389° N and longitude 99.9647083° E.The qualitative comparison 5 shows that U-Net with the EfficientNet B7 backbone produces sharper and

Table 3. Quantitative results of land cover segmentation across eight classes using U-Net and U-Net++ with different EfficientNet backbones. The results are reported as MIoU (%).

Model	Backbone	Bareland	Grass	Pavement	Road	Tree	Water	Cropland	Building	MIoU (%)
U-Net	EfficientNet B5	31.58	47.63	45.77	55.49	62.06	32.42	54.88	74.98	53.90
U-Net	EfficientNet B6	**39.31**	47.53	47.24	**57.96**	62.34	**38.82**	61.44	73.75	55.74
U-Net	EfficientNet B7	37.45	**49.28**	**48.76**	54.23	**63.66**	36.15	**65.65**	**78.14**	**56.53**
U-Net++	EfficientNet B6	17.19	36.06	37.73	39.19	53.98	15.49	38.97	48.23	39.30
DeepLabv3	EfficientNet B6	25.93	43.96	45.37	52.38	60.09	32.83	57.94	72.39	52.26

more consistent object boundaries compared to the models using EfficientNet B6. This difference becomes especially clear in areas with complex spatial patterns. For example, the B7 model is better at continuously separating forested regions from agricultural fields, detecting narrow rural roads and small buildings more accurately, and maintaining the continuity of irregular land use patterns such as riverbanks and mixed farmland areas.

These observations align with the quantitative results, where the B7 model achieved higher overall accuracy. The visual clarity in its predictions reflects its stronger ability to learn fine grained spatial features.

In contrast, U-Net with EfficientNet B6 shows more boundary blurring and a higher rate of misclassification along class transitions, particularly in regions where spectral characteristics change gradually, such as river edges and field boundaries. This behavior corresponds to the model's lower quantitative performance.

Overall, the qualitative results in Figure 3 reinforce the quantitative findings, demonstrating that using EfficientNet B7 as the backbone allows the model to deliver more accurate land-cover segmentation, with better spatial continuity, clearer boundaries, and fewer errors between visually similar classes.

4.2. Performance Analysis

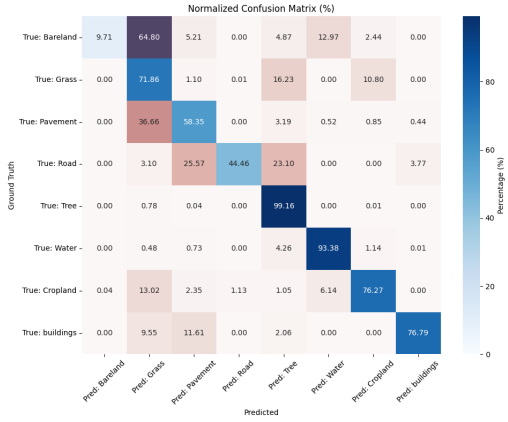


Fig. 6. Normalized Confusion Matrix (%) of the U-Net (EfficientNet B7) model.

To gain a deeper understanding of how well the model performs across individual classes, this study generated a Normalized Confusion Matrix (%) for the best performing architecture, U-Net with EfficientNet B7, as shown in Figure 6. The results highlight that the model classifies highly distinctive classes with remarkable accuracy. For example, Trees (99.6%), Water (99.1%), and Building (98.6%). These classes have strong visual characteristics in terms of color, texture, and geometric structure, which makes them easier for the model to correctly identify.

In the confusion matrix, darker red tones represent higher misclassification rates. Although the model performs accurately for most classes, some confusion remains, especially between Pavement and Road, which share similar spectral signatures and textural patterns. This leads to noticeable prediction overlap between the two classes. A moderate level of confusion is also observed between Grass and Cropland, which often exhibit similar reflectance patterns and physical characteristics, particularly during certain seasons. As a result, the boundaries between these two classes can appear ambiguous in satellite imagery.

Despite these misclassifications, the strong blue diagonal line indicates that U-Net (EfficientNet B7) maintains consistently high accuracy across all eight classes. This pattern reflects the model's ability to capture spatial context and extract multi level features effectively. Overall, while certain class pairs naturally exhibit overlap due to similarities in real landscapes, the EfficientNet B7 backbone demonstrates strong generalization capabilities and delivers superior performance compared to the other evaluated models.

5. Discussion

The experimental results of the U-Net and U-Net++ models with EfficientNet backbones reveal both strengths and limitations in performing land cover segmentation on high resolution satellite imagery.

5.1. Performance Highlights

As shown in Table 3, the U-Net B7 model achieved the best performance across several classes, particularly Tree, Cropland, and Building, with mIoU values reaching 65.65% for Cropland and 78.14% for Building. These classes performed well mainly because of their clear spatial patterns and continuity, which make it easier for the model to learn consistent pixel level features. Visual analysis 5 also shows that U-Net B7 accurately defines object boundaries, especially in urban and mixed urban-rural areas. This finding aligns with Hermosilla et al. (2022), who noted that land cover types with continuous structures and spectral signatures enable deep learning models to achieve higher segmentation accuracy.

5.2. Observed Challenges

Although U-Net with EfficientNet B7 achieved the highest overall performance with an mIoU of 56.53%, certain classes remain difficult for the model , particularly those with high visual similarity to other categories or those with limited training samples. This issue is most evident in the Bareland and Water classes, where the model achieved relatively low mIoU scores of 37.45% and 36.15%, respectively. Even though the model is strong in learning, the spectral similarity between Bareland and Pavement, as well as the overlapping color tones between Water and some regions of Grass, leads to frequent misclassification [13, 16].

The limitations become more evident when examining the performance of U-Net with EfficientNet B6. Although this model stands as one of the stronger baselines in the study, it still shows noticeable drops in performance for classes with irregular patterns or limited sample diversity. For instance, while U-Net B6 achieves a solid overall mIoU of 55.74%, its accuracy on Bareland 39.31% and Water 38.82% are lower compared with visually distinct classes such as Building or Cropland. This shows that even a well performing architecture can have difficulty generalizing when class appearance varies widely across different scenes or when the training samples do not fully capture that variation.

In summary, while EfficientNet B7 provides the best results for multi class, high resolution land cover classification, two major challenges remain: (1) classes that are spectrally similar to others, and (2) classes with too few training examples. Both factors significantly affect model performance and should be carefully addressed in future work, either by improving the dataset or by designing models that are more robust to these limitations.

5.3. Class-specific analysis

Table 4. Class-specific accuracy analysis highlighting performance differences across land cover types based on sample availability, spatial continuity, and feature distinctiveness.

Class	Accuracy	Factors
Tree, Cropland, Building	High	Clear patterns, spatial continuity, sufficient samples
Grass, Pavement, Road	Medium	Similar features, some confusion between classes
Bareland, Water	Low	Few samples, similar textures, unclear or inconsistent labels

The table and visual observations suggest that the model's performance is heavily influenced by how many samples each class has, how continuous their patterns are, and how distinct their features appear. Similar issues have been noted in earlier studies [16, 15], where open data often introduced challenges like unbalanced class distributions, inconsistent labeling, and domain shifts all of which can lower accuracy for certain classes.

6. Conclusion

In summary, this research evaluated the possibilities of generating a high-resolution land cover map for Chiang Rai province using open source datasets and modern Deep Learning architectures. We trained the models using an open dataset available on the Hugging Face platform and evaluated their performance using high resolution RGB images from Google Earth Pro, which represent the real world data of the target area.

The experimental results that U-Net utilizing EfficientNet B7 as the backbone yielded the best performance, achieving a mean Intersection over Union (mIoU) of 56.53% across all eight classes.

However, there are significant limitations that need to be addressed, namely the issue of domain shift between the training dataset and the target area, and class imbalance, which led to lower accuracy for classes with fewer samples.

In conclusion, this work shows that the approach of using open data combined with modern segmentation models can provide practically useful results for local planning and research, while significantly reducing data acquisition costs.

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